

IMAGINARY ELECTRIC COMPANY

Power Generation Division - Plant Integration

Alpilla 600 MW Combined Cycle Power Plant - 1 x IE-9H.02 multi-shaft power island

Power Island Thermal Performance Data (Heat Balance)

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CASE STUDY - Imaginary Electric, its products and all data are fictional.

1. Purpose and basis

This document gives the expected thermal performance of the IEC power island (1 x IE-9H.02 gas turbine with generator, 1 x IE-HR3 triple-pressure reheat HRSG, 1 x IE-ST3R reheat steam turbine with generator, seawater condenser) for the EPC's plant heat and mass balance. Values are **expected values, new and clean**, without guarantee margins; IEC's guarantee values to the consortium are given in section 7.

- Case **SRC-NG-100**: Site Reference Conditions per ER-02.03 - ambient 15.0 degC, 70 % RH, 1011.5 mbar; seawater 16.0 degC; base load on natural gas; power factor 0.85 lagging; 50.0 Hz.
- Fuel: Owner reference composition ALP-OWN-FUL-001 / ER-03.02. **LHV calculated per ISO 6976 from this composition: 47.48 MJ/kg (34.78 MJ/Sm3, 15 degC/15 degC)**. Heat input = fuel mass flow x this LHV. Fuel gas heated to 215 degC in the IEC performance gas heater with IP feedwater.
- GT inlet pressure loss 10 mbar, exhaust back-pressure 36 mbar (HRSG 30 mbar + stack 6 mbar, stack by EPC).
- Steam piping between HRSG and steam turbine is **by EPC**; IEC assumed pressure drops HP 7.0 bar, hot reheat 1.6 bar, cold reheat 0.8 bar, LP 0.45 bar, and temperature drops of 3 K (HP, HRH), 1 K (CRH), 2 K (LP). The EPC shall confirm or IEC will correct the performance.
- Boiler feed pumps and condensate pumps are by EPC; IEC assumed BFP discharge 209 bar(a) (HP) / 46 bar(a) (IP), CEP discharge 16 bar(a).
- Steam turbine gland leak-offs, valve stem leakages and HRSG blowdown are not shown in this summary heat balance (blowdown zero at design; leak-offs are internal to the ST).

2. Summary at SRC-NG-100

Item	Value	Unit
GT output at generator terminals	425.00	MW
GT heat rate (LHV)	8,314.1	kJ/kWh
GT heat input (LHV)	981.52	MW
Fuel gas flow	20.672 kg/s = 101,596	Sm3/h
Compressor inlet air flow	776.01	kg/s
GT exhaust flow / temperature	796.68 / 640.0	kg/s / degC
GT generator efficiency	98.90	%
GT losses other than generator (lube oil, radiation)	3.80	MW
Performance gas heater duty	9.46	MW
HRSG heat transferred to water/steam	500.83	MW
HRSG heat loss (casing, % of gas-side duty)	0.4	%
Stack temperature	80.5	degC
ST output at generator terminals	200.31	MW
ST mechanical / generator efficiency	99.55 / 98.80	%
Condenser pressure	34.44	mbar(a)
Condenser duty	289.96	MW
Circulating water flow (7 K rise)	10,179 kg/s = 36,050	m3/h
Power island gross output	625.31	MW
Power island gross heat rate (LHV)	5,650.8	kJ/kWh

Net output and net heat rate depend on the EPC balance-of-plant auxiliary loads and transformer losses; IEC's auxiliary loads are listed in IEC-ALP-REQ-001 section 7.

3. Stream data at SRC-NG-100

Node codes: AMB ambient, FGS EPC fuel gas system, FGH IEC performance gas heater, GT gas turbine, HRSG, STK stack (EPC), PIPE steam piping HRSG-ST (EPC), ST steam turbine, COND condenser, CEP condensate pumps (EPC), CMIX condensate/FGH-return junction (EPC), BFP boiler feed pumps (EPC), SEA intake, OUT outfall.

No	Description	Fluid	From	To	Flow kg/s	p bar(a)	T degC	h kJ/kg
1	Ambient air to GT compressor inlet	air	AMB	GT	776.01	1.01	15.0	-
2	Fuel gas at performance gas heater inlet (IEC terminal point)	fuel_gas	FGS	FGH	20.67	38.00	25.0	-
3	Heated fuel gas to GT combustion system	fuel_gas	FGH	GT	20.67	35.50	215.0	-
4	GT exhaust gas to HRSG inlet	flue_gas	GT	HRSG	796.68	1.05	640.0	-
5	Flue gas at HRSG outlet to stack	flue_gas	HRSG	STK	796.68	1.01	80.5	-
6	HP steam at HRSG outlet (HP SH outlet)	steam	HRSG	PIPE	100.16	187.00	600.0	-
7	HP steam at ST main stop valve inlet	steam	PIPE	ST	100.16	180.00	597.0	-
8	Cold reheat steam at HP turbine exhaust	steam	ST	PIPE	100.16	40.20	366.2	-
9	Cold reheat steam at HRSG reheater inlet	steam	PIPE	HRSG	100.16	39.40	365.2	-
10	IP superheated steam (joins cold reheat inside the HRSG)	steam	HRSG	HRSG	17.34	40.30	330.0	-
11	Hot reheat steam at HRSG outlet	steam	HRSG	PIPE	117.50	37.60	600.0	-
12	Hot reheat steam at ST IP stop/control valve inlet	steam	PIPE	ST	117.50	36.00	597.0	-
13	LP steam at HRSG outlet	steam	HRSG	PIPE	11.39	5.40	300.0	-
14	LP steam at ST LP admission	steam	PIPE	ST	11.39	4.95	298.0	-
15	IP turbine exhaust (crossover, internal)	steam	ST	ST	117.50	4.95	308.8	-
16	LP turbine exhaust to condenser (UEEP)	steam	ST	COND	128.89	0.0344	26.4	2,360.4
17	Condensate at hotwell outlet / CEP suction	water	COND	CEP	128.89	0.0344	26.3	-
18	Condensate at CEP discharge	water	CEP	CMIX	128.89	16.00	26.5	-
19	Fuel gas heater water return to condensate	water	FGH	CMIX	12.48	43.00	75.0	-
20	Condensate to HRSG condensate preheater inlet	water	CMIX	HRSG	141.37	15.50	30.9	-
21	Feedwater from LP drum to boiler feed pumps	water	HRSG	BFP	129.98	5.90	158.1	-
22	HP feedwater at BFP discharge	water	BFP	HRSG	100.16	209.00	161.8	-
23	IP feedwater at BFP interstage outlet	water	BFP	HRSG	29.82	46.00	158.9	-
24	IP economiser water to performance fuel gas heater	water	HRSG	FGH	12.48	43.30	248.0	-
25	Circulating seawater to condenser inlet	seawater	SEA	COND	10,179.10	2.50	16.0	-
26	Circulating seawater from condenser outlet	seawater	COND	OUT	10,179.10	1.70	23.0	-

4. GT exhaust gas composition at SRC-NG-100 (mol %, wet)

N2	O2	Ar	CO2	H2O
73.85	11.12	0.88	4.47	9.67

5. HRSG temperature profile at SRC-NG-100

HRSG section (gas flow order)	Gas in degC	Gas out degC	Duty MW
HPSH / RH	640.0	450.6	176.83
HPEVAP	450.6	371.6	71.39
HPECO2	371.6	299.9	63.71
IPSH	299.9	295.1	4.19
IPEVAP	295.1	261.0	29.89

HRSG section (gas flow order)	Gas in degC	Gas out degC	Duty MW
HPECO1 / IPECO	261.0	201.5	51.64
LPSH	201.5	197.4	3.51
LPEVAP	197.4	166.2	26.83
CPH	166.2	80.5	72.85

Pinch points HP/IP/LP: 8 / 8 / 8 K; approach temperatures at the economiser outlets: 5 / 5 / 5 K. Drum saturation temperatures HP/IP/LP: 363.6 / 253.0 / 158.2 degC. Steam turbine section outputs (shaft): HP 41.68 MW, IP 69.13 MW, LP 92.85 MW; LP exhaust moisture 7.74 %.

6. Performance at other ambient conditions and part load (estimates)

Base load on natural gas, seawater temperature as shown, other conditions as SRC. GT from the IE-9H.02 correction curves; the bottoming cycle is re-estimated with the design pinch/approach values. **These are preliminary estimates, not off-design ratings of the fixed HRSG and condenser surfaces**; rated off-design heat balances follow with the HRSG thermal design (IEC-ALP-DS-104 Rev A).

Ambient degC	RH %	Seawater degC	GT MW	GT HR kJ/kWh	Exhaust kg/s	Exhaust degC	ST MW	Gross MW	Gross HR kJ/kWh	Condenser mbar	Gas Sm3/h	Remark
-8.0	80	6.9	470.0	8,089	823.5	634.0	210.6	680.6	5,586	19.8	109,310	GT at output limit 470 MW
-4.8	80	8.0	470.0	8,131	832.5	635.7	213.0	683.0	5,595	21.2	109,874	GT at output limit 470 MW
5.0	75	11.0	452.8	8,217	829.2	634.8	209.3	662.1	5,620	25.5	106,984	
15.0	70	16.0	425.0	8,314	796.7	640.0	200.3	625.3	5,651	34.4	101,596	
25.0	55	22.0	397.2	8,411	761.8	645.2	190.0	587.2	5,689	48.7	96,053	
32.8	45	24.0	372.8	8,501	729.5	649.3	182.9	555.7	5,703	54.4	91,121	
40.0	35	27.0	348.2	8,595	695.9	653.0	174.3	522.5	5,728	64.2	86,055	

GT output is limited to **470 MW** at the generator terminals (generator and shaft limit) below about -1 degC; the GT then runs at part load with the IGV closing.

GT load %	GT MW	GT HR kJ/kWh	Exhaust kg/s	Exhaust degC	ST MW	Gross MW	Gross HR kJ/kWh
100	425.0	8,314	796.7	640.0	200.3	625.3	5,651
80	340.0	8,746	696.5	640.0	174.8	514.8	5,776
60	255.0	9,436	593.1	640.0	148.5	403.5	5,963
50	212.5	9,894	541.0	632.0	132.4	344.9	6,096
40	170.0	10,600	496.9	615.0	115.7	285.7	6,307

Part load at SRC with the steam cycle at fixed pressure (sliding-pressure operation will be optimised in detailed design; values are conservative estimates).

7. IEC guarantees to the consortium (power island)

Guarantee (SRC-NG-100, new and clean, per ASME PTC 46 / ISO 2314)	Value
Power island gross output at generator terminals	not less than 614.0 MW
Power island gross heat rate (LHV)	not more than 5,745 kJ/kWh
GT output at generator terminals	not less than 418.0 MW
Condenser pressure at 16.0 degC seawater, 36,050 m3/h	not more than 35.5 mbar(a)
IEC auxiliary loads (IEC-ALP-REQ-001 table 7.1, excl. GSU losses)	not more than 2,700 kW
NOx / CO at HRSG outlet (15 % O2, dry), 40-100 % GT load, natural gas	30 / 30 mg/Nm3

Degradation is excluded from the values above; recoverable and non-recoverable degradation curves follow in IEC-ALP-PER-002.